Dayananda Sagar University Bengaluru

Section C

Title: AI - Based Load Forecasting for Balancing Demand and Supply

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TITLE OF PROJECT:

AI - Based Load Forecasting For Balancing Demand and Supply

Objectives

- To accurately predict electrical current using Machine Learning models based on Power, Voltage, and Power Factor.

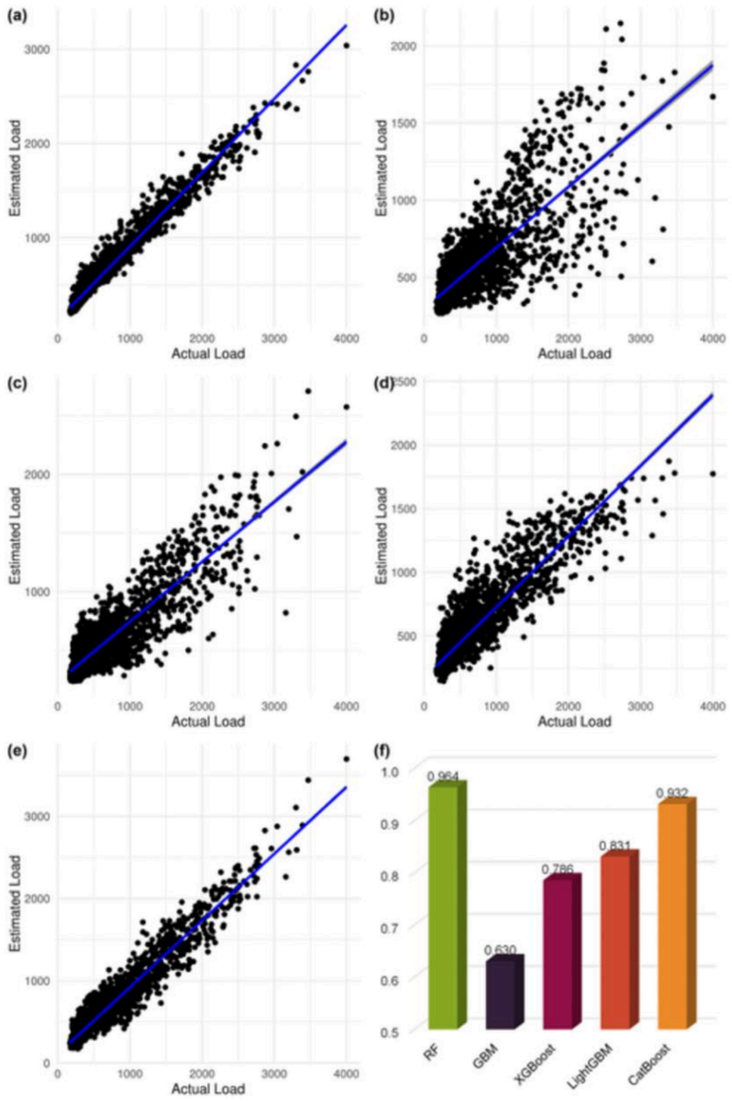
- To develop a user-friendly GUI system that provides instant current forecasting and supports efficient energy usage aligned with SDG 7.

Introduction

- Electricity demand changes constantly, and inaccurate forecasting can lead to overloads, power cuts, and energy wastage.

- AI-based forecasting helps understand consumption patterns and supports smarter, more sustainable energy management aligned with SDG 7.

Literature Review



Scatterplots and R2 comparison for each decision tree–ensemble learning model.
(a) random forest
(b)gradient boosting machine
(c) extreme gradient boosting
(d) light gradient boosting machine
(e) categorical boosting
(f) R2 comparison.

Literature Review

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Models	$MAPE_W$	$MAPE_H$	$MAPE_T$	$CVRMSE_W$	$CVRMSE_H$	$CVRMSE_T$	$NMAE_W$	$NMAE_H$	$NMAE_T$
RF	11.14	12.37	11.54	14.33	16.83	15.05	11.64	13.16	12.07
GBM	11.43	12.42	11.75	14.48	16.70	15.12	11.90	13.19	12.27
XGBoost	11.44	12.78	11.88	14.61	17.32	15.39	11.86	13.50	12.33
LightGBM	11.25	12.45	11.64	14.24	16.72	14.95	11.70	13.20	12.13
CatBoost	11.49	12.66	11.87	14.63	17.02	15.32	11.92	13.38	12.34

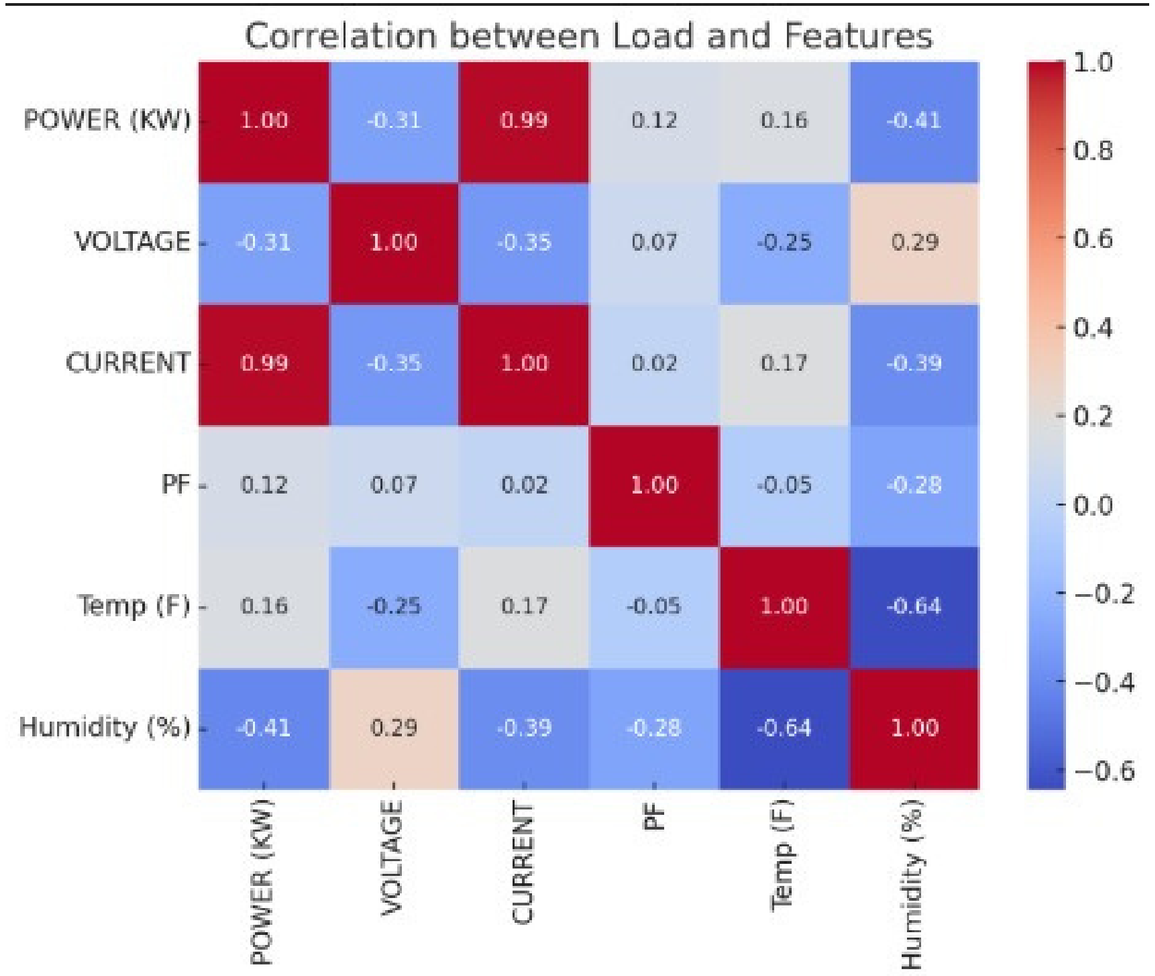
Comparative analysis of decision tree–based ensemble learning models. All evaluation metrics are presented in percentage (%).

Dataset

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DATE	TIME	VOLTAGE	CURRENT	PF	POWER (KW)	*WEEKEND/WEEKD*	SEASON	Temp (F)	Humidity (%)	Substation Shutdown
1/1/2021	01-00	11.6	102	0.96	1967	0	1	65	90	0
	02-00	11.6	102	0.96	1967	0	1	65	90	0
	03-00	11.6	102	0.96	1967	0	1	65	90	0
	04-00	11.3	130	0.96	2443	0	1	78	49	0
	05-00	11.2	148	0.96	2756	0	1	78	49	0
	06-00	11.2	172	0.96	3203	0	1	78	49	0
	07-00	11.5	218	0.96	4169	0	1	83	34	0
	08-00	11.2	274	0.96	5103	0	1	83	34	0
	09-00	11.2	280	0.96	5214	0	1	83	34	0
	10-00	11.5	254	0.96	4857	0	1	81	42	0
	11-00	11.5	254	0.96	4857	0	1	81	42	0
	12-00	11.5	243	0.96	4647	0	1	81	42	0
	13-00	11.6	231	0.96	4456	0	1	71	42	0
	14-00	11.8	214	0.96	4199	0	1	71	87	0
	15-00	11.8	205	0.96	4022	0	1	71	87	0
	16-00	11.7	215	0.96	4183	0	1	67	87	0
	17-00	11.5	194	0.96	3710	0	1	67	87	0
	18-00	11.7	44	0.96	856	0	1	67	87	0
	19-00	11.8	42	0.96	824	0	1	66	88	0
	20-00	11.8	39	0.96	765	0	1	66	88	0
	21-00	11.8	30	0.96	589	0	1	66	88	0
	22-00	11.8	25	0.96	491	0	1	62	89	0
	23-00	11.8	22	0.96	432	0	1	62	89	0
	24-00	11.8	21	0.96	412	0	1	62	89	0
	01-00	11.7	95	0.95	1829	0	1	65	90	0
	02-00	11.7	95	0.95	1829	0	1	65	90	0
	03-00	11.7	95	0.95	1829	0	1	65	90	0
	04-00	11.5	100	0.95	1892	0	1	78	48	0
	05-00	11.4	135	0.95	2532	0	1	78	48	0
	06-00	11.3	162	0.95	3012	0	1	78	48	0
	07-00	11.5	213	0.95	4031	0	1	83	32	0
	08-00	11.3	257	0.95	4779	0	1	83	32	0

Dataset



Features:

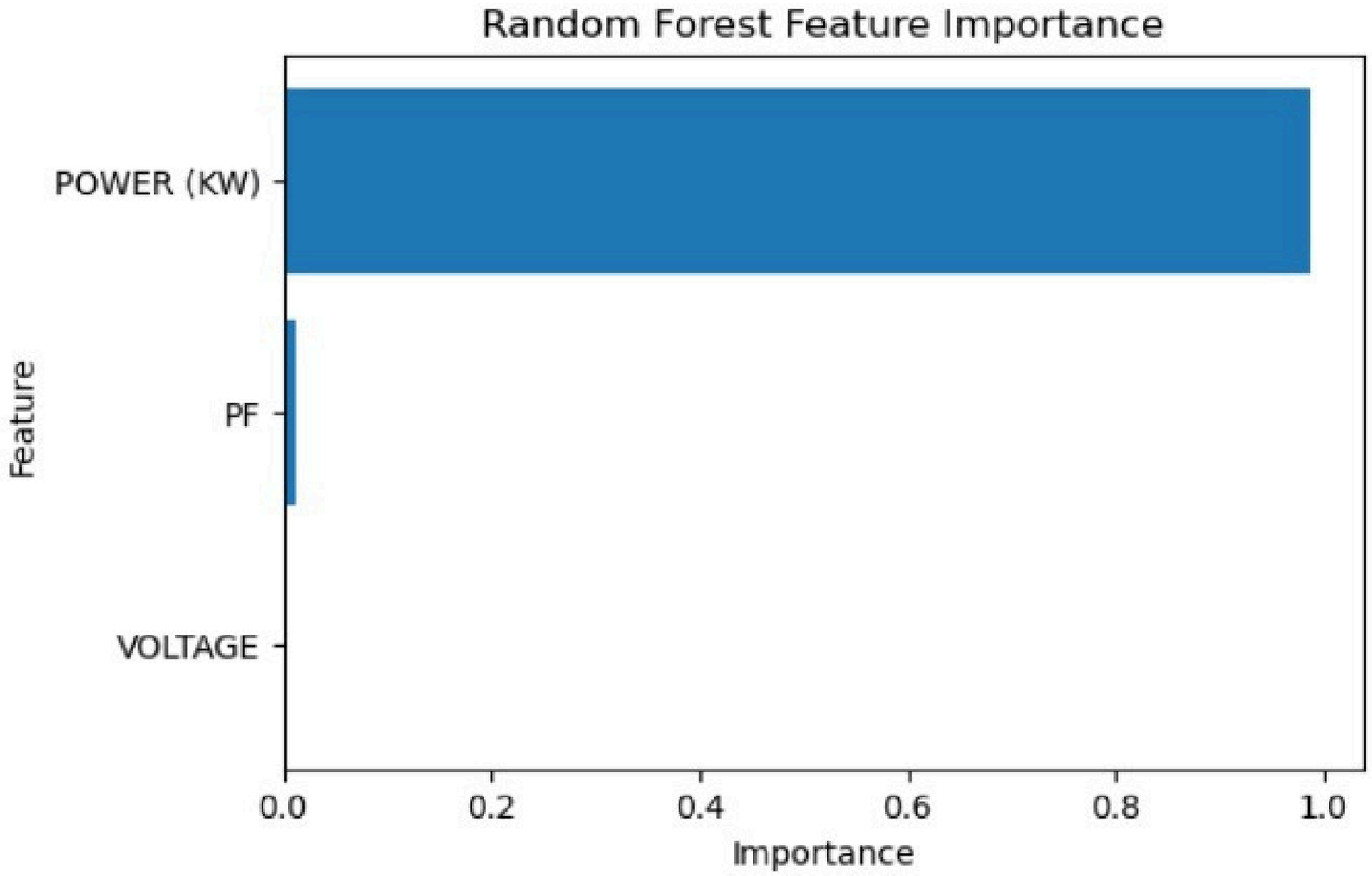
1) Electrical: Voltage, Current, PF, Power

2) Calender: Weekday/Weekend, Season.

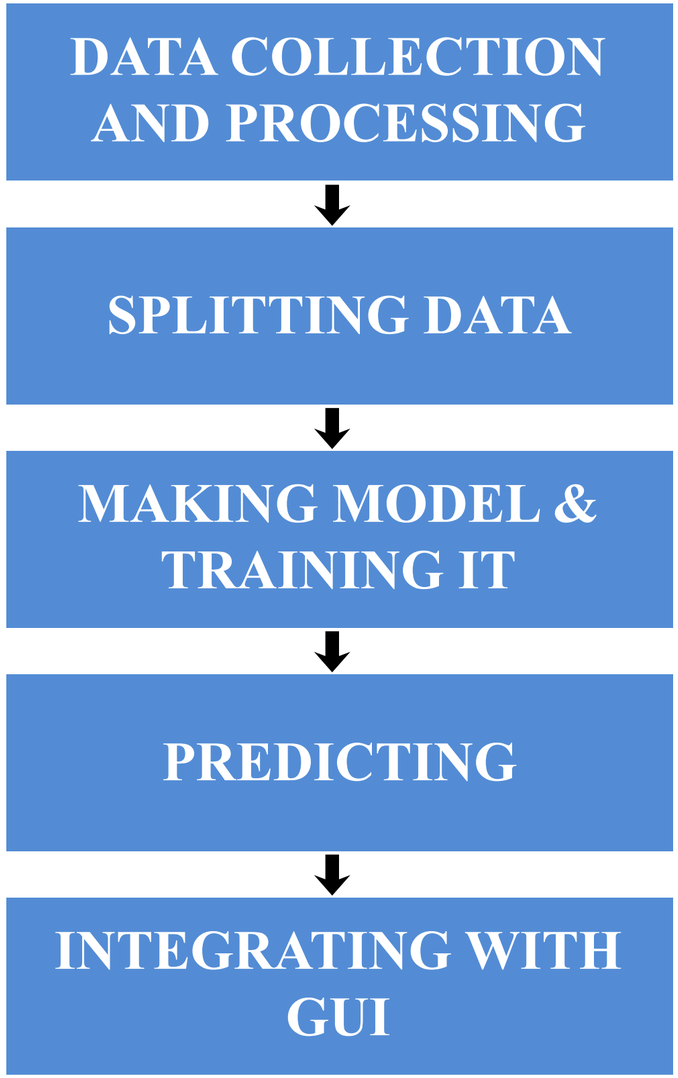
3) Weather: Temperature, Humidity

Target Variable = CURRENT

Dataset

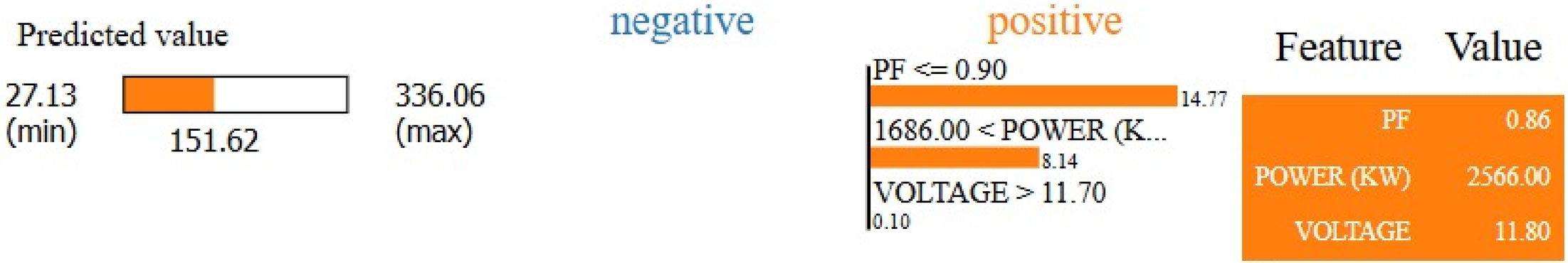

Random Forest Feature Importance
POWER (KW)
PF
VOLTAGE
Feature
0.0
0.2
0.4
0.6
0.8
1.0
Importance

Methodology

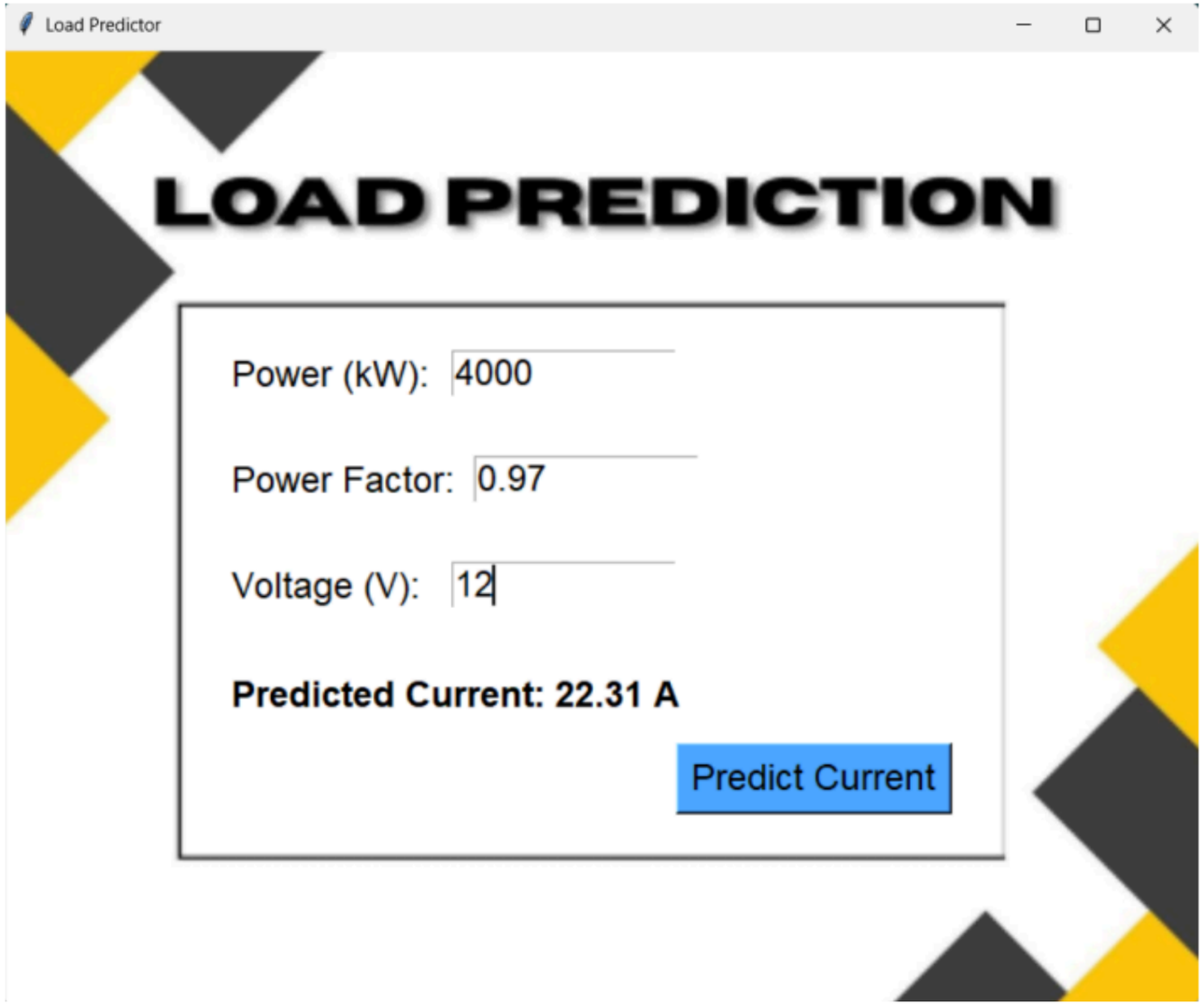


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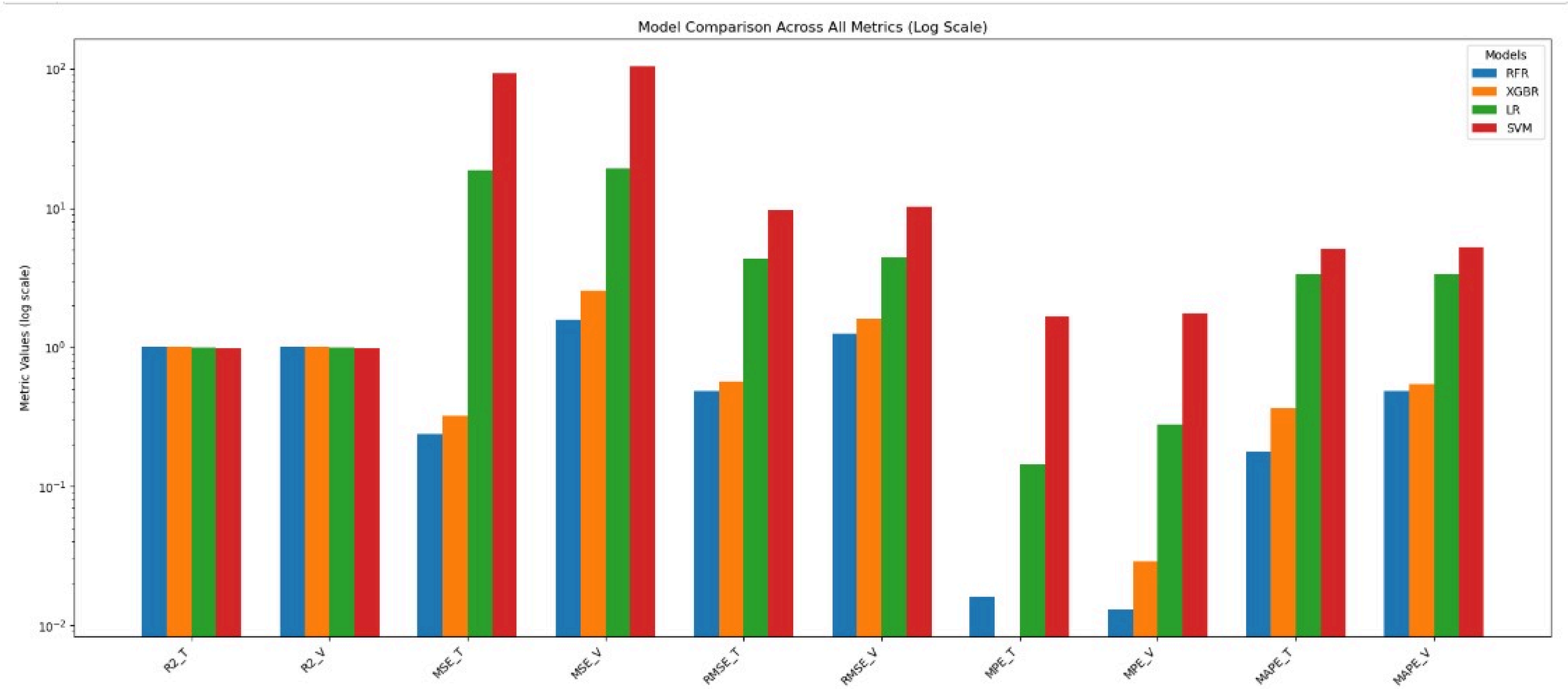
LIME



Results


Load Predictor
LOAD PREDICTION
Power (kW): 4000
Power Factor: 0.97
Voltage (V): 12
Predicted Current: 22.31 A
Predict Current

Conclusions




Thank You